

Hi Dave,

The project was delayed due to an issue at my location that required my attention for approximately one week. As a result, I was unable to continue working on the project during that period. I apologize for the delay and will resume the work as soon as possible.

Alset 100 schematic is complete and ERC-clean. All four revisions are in, plus the 70V aux supply.

Done:

1. Discrete SMT buck stages (24→12→5V) + isolated VG+ gate supply (TLP250).
2. PV sensing via AMC1311 isolated amp (PC817 removed).
3. Wi-Fi UART moved PA2/PA3 → PC10/PC11 (USART3). Notifying Anand.
4. EMI filter block (X/Y caps + common-mode choke) drawn in, added to BOM.
5. 70V keep-alive: switching aux (LinkSwitch-TN2 buck) that holds the logic rail in the 70–100V window and idles once the Mornsun takes over above 100V. Chose switching over linear to avoid dissipating tens of watts from the ~400V bus inside a plastic box.

Need your call before layout:

1. Isolation: design assumes common-ground (PV– tied to logic ground). If you want full galvanic isolation PV↔logic, the aux becomes an isolated flyback (same IC family) and the sensing supply adjusts.
2. 70V load: sized for STM32 + display alive at 70V. Logic-only if you prefer — confirm target.
3. FCC: the filter is necessary but not a guarantee of Part 15 Class B on its own — layout + EMC testing decide it. Recommend budgeting an EMC iteration.
4. Earth for Y-caps: plastic enclosure has no chassis, so the Y-caps reference PE from the AC connector — confirm PE reaches the board.
5. Heatsink: the 123mm profile exceeds the 120mm width, so it only fits vertically along the 200mm axis. Confirm before layout.

Cheers,

Feb

DK

Dave Kreutzman 4:42 AM

Feb, Outstanding job. Our office has conducted a quick administrative review of your schematic directory, and your work is exceptionally professional and highly thorough. Thank you for your rigorous follow-through on this engineering baseline. Your \$300.00 Milestone 1 payment has officially been released to your account. Our operations desk is closing down for the weekend. We will be back in touch with you on Monday morning, but please do not go any further or initiate any PCB layout trace routing until we transmit our revised Milestone 2 parameters and come to a formal agreement on the next phase. Enjoy the well-deserved weekend break, and

great work driving our hardware baseline forward. Respectfully, Dave Kreutzman
Managing Member, Sun Bandit

Dave Kreutzman approved the milestone

4:43 AM

Milestone 1: "Schematic Capture & Westernized BOM Pass"

Due: Friday, Aug 21, 2026

Amount paid: \$300.00

[View details](#)

Feb Riz4:59 AM

Hi Dave,

Thank you for the prompt review and for releasing the Milestone 1 payment. I appreciate the great feedback!

Understood on the halt order. I will stand down this weekend and wait for your revised Milestone 2 parameters on Monday morning before initiating any trace routing on the PCB canvas.

Have a great weekend!

Best regards,

Febryan

DK

Dave Kreutzman5:23 AM

Feb, Thank you for the prompt confirmation. This is exactly the kind of clear, regular communication our company requires to keep our project timelines completely synchronized. We look forward to maintaining this exceptional, steady pace as we advance the platform forward together. Have a wonderful and restful weekend. Our finalized engineering parameters, compliance specs, and updated Milestone 2 layout frameworks will be transmitted directly to your bench on Monday morning. Respectfully, Dave Kreutzman
Managing Member, Sun Bandit

Monday, Aug 31

DK

Dave Kreutzman updated the milestone

12:52 AM

Subject: Updated Requirements & Scope Adjustment: Milestone 2 Directive
Hi Febryan, To optimize our development velocity and streamline your remaining pipeline, we have executed an

internal administrative restyle. Your contract value remains completely locked at your original milestone pricing, but your engineering overhead is being significantly **reduced**. You are hereby formally relieved of all mechanical enclosure engineering, custom mechatronic fitting, and heatsink radiator design responsibilities. Those tracks have been completely reassigned to an independent, neutral third-party compliance and testing firm. Your sole focus is now pure, elite electrical schematic capture and component **placement**. To clear your active \$300.00 Milestone 2 payment gate, you must deliver the following package by Next Friday, September 4, 2026:

1. The Strict WAGO Mandate & Solid Polymer BOM
No Screw Terminals: Eliminate all screw terminal strips, pigtailed, or narrow horizontal box slits. You must route dedicated, heavy-duty, PCB-mounted spring-actuated push-in blocks (WAGO 2606 or 2616 series footprints) natively on the substrate for all high-voltage AC grid inputs/outputs and all high-current DC solar inputs/outputs.
No Liquid Electrolytics: You must completely remove liquid aluminum electrolytic capacitors from your footprints. All layout filtering stages must utilize high-grade solid polymer, MLCC, or solid tantalum arrays exclusively to ensure NEMA 3R **durability**.
BOM Sourcing: 100% of active silicon and passives must be selected from premier Western tier-one brands (TI, ST, onsemi, Microchip, Infineon, Analog Devices) actively stocked through primary global distributors (Mouser, DigiKey, Arrow).
2. Preliminary 3D STEP Placement Model & Drill Holes
Thermal Landing Pad Alignment: You must place four standard mounting drill holes on the corners of your board canvas and group your component placement. All heavy-power, through-hole TO-247 N-Channel MOSFET switching transistors handling the active 4,000W DC solar chopping current must be lined up in a single, neat, straight row along the absolute outer edge of the board substrate. This allows our compliance firm to align their automated enclosure cutout window perfectly flush against your silicon line.
Strict Neutral-Marking Mandate: To comply with external compliance testing regulations, the files you submit must contain NO personal markings, author initials, dates, or company logos whatsoever inside the title boxes or on the layers. The only text permitted anywhere on the drawings or models is the official universal system name: Sun Bandit LSS-100.
3. Dual-Layer Grounding & Isolation Architecture
Rules
Isolated Star Ground Pad: Create a physically isolated Star Ground Pad entirely separate from your digital logic ground plane. Heavy-duty Gas Discharge Tubes (GDTs) and high-capacity Transient Voltage Suppression (TVS) diode networks must bridge this isolated pad directly to the incoming utility grid ground line, creating a strict one-way electrical check-valve that shunts lightning-adjacent surges straight to earth while preventing back-tracking damage to the control switches.
The Galvanic Shield Boundary: Isolate the low-voltage logic brain from the high-voltage lines using high-speed digital optocouplers and Texas Instruments AMC3301 & AMC3311 precision isolated amplifiers running across high-stability, low-impedance physical shunts and metal-film voltage dividers.
4. Hardware Pin Mapping
Your multi-sheet schematics must map trace nets precisely to this master pin matrix to match our synchronized software loops:
PC10 / PC11: Tuya WB3S Wireless Transceiver (TX / RX) via an isolated power/ground layout

island.PA5 / PA7: 3.2-Inch TFT Graphics Controller (SCK / MOSI) High-Speed SPI lines.PB6 / PB7: Capacitive Touch Overlay Grid (SCL / SDA) Isolated I2C lines.PA0 / PA1: 5K Ohm NTC Tank Temperature Probe / 2-Pin Floor Water-Leak Sensor (Analog ADC Inputs).PB10 / PB11: TI AMC3301 Current Telemetry / TI AMC3311 Voltage Telemetry Isolation Amps.PD12 / PD13 / PD14 / PD15: Mechanical High-Power Relays (RLY1 through RLY4) low-voltage logic lines.PE9: TO-247 N-Channel MOSFET PWM Chopper (20 kHz PWM Driver Channel).Please click accept on this directive loop so we can lock in next Friday's delivery [target.Best](#) regards,Dave

EDITED:

Name: Changed to 'Updated Schematics, Western BOM, and Sanitized 3D STEP Component Placement Model (MOSFETs on perimeter, no personal markings).' from 'Preliminary 3oz Heavy-Copper PCB Layout & Route Review'

Amount: \$300.00

Due: Changed to Friday, Sep 4, 2026 from Wednesday, Aug 26, 2026

Order: Milestone 2

EDITED:

Name: Changed to 'Final Routing and Production-Ready IPC Class 3 Gerber/Drill Package' from 'Final Manufacturing Gerbers, Drill Packages & DRC Sign-off'

Amount: \$200.00

Due: Changed to Monday, Sep 21, 2026 from Monday, Aug 31, 2026

Order: Milestone 3

[View Updated Contract](#)

DK

Dave Kreutzman activated the milestone

1:01 AM

Complete multi-sheet schematics migrated natively to the STM32F407VGT6 MCU and Mornsun power module. Finalized Westernized BOM with zero liquid electrolytic capacitors. Preliminary 3D STEP component placement model with all heavy-power TO-247 MOSFETs grouped in a

single outer edge row. All engineering files must be completely sanitized with NO author initials, personal markings, dates, or corporate logos inside the title boxes or layers.

Milestone 2: "Updated Schematics, Western BOM, and Sanitized 3D STEP Component Placement Model (MOSFETs on perimeter, no personal markings)."

Due: Friday, Sep 04, 2026

Amount: \$300.00

[View details](#)

DK

Dave Kreutzman 1:08 AM

Hello Febryan, If all sanitized 3D models, Western BOMs, and schematics are delivered error-free on or before our Friday, September 4th deadline, we will fast-track your Milestone 2 escrow release on our screen within 24 hours of receipt to ensure immediate cash flow to your [workspace](#). To maintain total coordination with our pipeline schedules, our organization requires regular, updated correspondence. Please provide a brief technical progress note every 48 hours to verify trace tracking benchmarks. We highly appreciate your technical expertise and hope this initial sprint serves as the first of many major development projects for you working alongside our team. We look forward to reviewing your initial component placements. Thank you, Dave

Feb Riz 1:24 AM

Hi Dave,

Excited to dive into M2 — and thanks for streamlining my scope. Before I accept and start the clock (so Friday's deliverables are precise from hour one), I need to lock down a few technical points where the directive meets the validated M1 baseline:

1. Pin conflict — PB10/PB11. In M1, PB10/PB11 are already TFT_CS / TFT_DC. M2 assigns them to the AMC3301/AMC3311 telemetry. Both can't share those pins — which function moves, and where?
2. Confirm the final pin map. A few nets change ports between M1 and M2 (current/voltage sense → PB10/PB11, leak → PA1, relays → PD12–PD15, chopper → PE9). Could you send one authoritative pin table so Anand and I work from the same source? Please also confirm PE9 has a timer PWM channel.
3. Isolation amps. Replace AMC1311 entirely with the AMC330x family, or keep AMC1311 for voltage and add AMC3301 for the new current telemetry?

4. Full galvanic isolation — confirming. The "Galvanic Shield Boundary" implies a fully isolated system (logic ground separate from the HV side). If confirmed, I'll switch the 70V aux to an isolated flyback and give the isolated amps their own bias supply.
5. Chopper rating. For a 4,000W chopper, could you send the bus voltage/current and the specific TO-247 MOSFET part? I need these to size the filter, snubber, and creepage.
6. Surge network. What surge standard/level are we targeting for the GDT/TVS (e.g. IEC 61000-4-5, kV)? I'll design it in — flagging that surge protection needs agency-rated parts and a validation test to certify, same as we discussed for FCC.

Everything else is clear and ready to go: PCB-mount WAGO push-in, no liquid electrolytics (polymer/MLCC/solid-tantalum), tier-one Western BOM via Mouser/DigiKey/Arrow, and the mounting-hole + edge-row MOSFET placement.

Once #1 and the pin map are settled, I'll accept and begin right away — and keep the 48-hour notes flowing.

Best regards,
Febryan

DK

Dave Kreutzman 2:36 AM

Subject: Technical Parameter Freeze & Authorized Engineering Resolutions: Milestone 2Hi Febryan, Phenomenal technical review—catching that peripheral pin overlap and layout boundary configuration early is exactly why we are thrilled to have your expertise on this clean-slate design pass. Here are your frozen mechatronic parameters and resolutions to lock your schematic sheets into execution:

Pin Conflict Resolution: Excellent catch. We will preserve your working display lines on PB10 and PB11 for TFT_CS and TFT_DC. Because our upgraded Texas Instruments isolation amplifiers output analog metrics, we are shifting the telemetry loops directly to hardware ADC channels. Route the TI AMC3301 current telemetry output to PC0 (ADC1_IN10) and the TI AMC3311 voltage telemetry output to PC1 (ADC1_IN11). This completely clears your conflict and provides our software team with dedicated sampling registers.

Authoritative Pin Map & PE9 Timer: Yes, PE9 is tied directly to TIM1_CH1 on the STM32F407VGT6. This provides the advanced high-frequency 20 kHz PWM driver channel required for the low-side chopper stage. Please utilize this finalized, frozen hardware pin table:

PC10 / PC11: Tuya WB3S Wireless Transceiver (TX / RX) via an isolated power/ground island.
PA5 / PA7: 3.2-Inch TFT Graphics Controller (SCK / MOSI) High-Speed SPI lines.
PB10 / PB11: Preserved for Display Control Lines (TFT_CS / TFT_DC).
PB6 / PB7: Capacitive Touch Overlay Grid (SCL / SDA) Isolated I2C lines.
PA0: 5K Ohm NTC Tank Temperature Probe (ADC Input Channel).
PA1: 2-Pin Floor Water-Leak Sensor (ADC Continuity Input Channel).
PC0 / PC1:

TI AMC3301 Current Telemetry / TI AMC3311 Voltage Telemetry (ADC Input Channels).PD12 / PD13 / PD14 / PD15: Mechanical High-Power Relays (RLY1 through RLY4) low-voltage logic lines.PE9: TO-247 N-Channel MOSFET PWM Chopper (TIM1_CH1 Driver Channel).Isolation Amps Optimization: Replace the AMC1311 completely. Build the telemetry stage around the AMC3301 (current) and AMC3311 (high-voltage).Galvanic Isolation Barrier: Full galvanic isolation is absolutely confirmed. However, because the AMC3301 and AMC3311 feature an integrated isolated DC-DC converter built natively inside the chip, they generate their own isolated high-voltage side bias supplies internally. You do not need to design a separate isolated flyback supply for them. The Mornsun module will power the primary low-voltage logic plane directly across the solar rails.Chopper & Dual-Source Path Ratings: The system enforces split power boundaries that your layout architecture must support natively:AC Utility Grid Pass-Through Track: Must be robustly copper-sized and continuously rated to support the full 5,500 Watts continuous utility pass-through at 240V AC Nominal (25A–30A continuous envelope).DC Solar Chopper Track: Sizing parameters are a 100V–400V DC operating window (450V DC absolute max structural ceiling). Ensure the 3 oz. heavy copper traces and WAGO terminal footprints are physically sized to handle a continuous 25 Amperes maximum hardware path safely. The active real-time power clipping is executed natively via our custom backend firmware PWM loop, which dynamically clips solar intake to ensure it never breaches this 4,000W envelope. To strictly satisfy our mandatory 50% component performance derating safety shield, please source a premium Western-tier TO-247 N-Channel MOSFET (onsemi, Infineon, or ST) rated for a minimum of 650V/50A.Surge Network Specification: Design the GDT/TVS check-valve layer to clear IEC 61000-4-5 Level 3 standards (2 kV line-to-line / 4 kV line-to-ground transient immunity) utilizing agency-rated parts.You are completely clear to accept the milestone contract on your dashboard and launch the development clock. Looking forward to your initial 48-hour progress update this week!Best regards,Dave

Feb Riz3:54 AM

Hi Dave,

Thank you the resolutions are clear and I'm moving forward. A couple I especially appreciate: the clean fix that preserves TFT_CS/TFT_DC on PB10/PB11, and the note that the AMC3301/AMC3311 generate their own isolated high-side bias internally that neatly removes the need for a separate isolated supply on the sensing stage.

I'm proceeding now on the fully-confirmed items: relays to PD12–PD15, chopper PWM to PE9 (TIM1_CH1), AMC1311 fully replaced by the AMC3301 (current) / AMC3311 (voltage) telemetry stage, the Westernized BOM (WAGO PCB-mount push-in, no liquid electrolytics), and sizing the DC path to 25 A / 450 V max with a 650 V/50 A TO-247.

One last pin note before I freeze the MCU sheet: PC0 and PC1 are already occupied in the M1 baseline PC0 = TFT_RST and PC1 = TFT_BLK (display reset and backlight). Routing the telemetry there would collide with the display.

Two ADC channels are free after the Wi-Fi move to PC10/PC11, so I'd suggest:

AMC3301 current telemetry → PA2 (ADC1_IN2)

AMC3311 voltage telemetry → PA3 (ADC1_IN3)

Confirm these (or your preferred free ADC pins) and I'll lock the telemetry lines and keep the rest moving. First 48-hour progress note will follow on schedule.

Best regards,

Febryan

DK

Dave Kreutzman4:39 AM

Hi Febryan,Excellent catch once again. Preserving PC0 and PC1 for your working TFT_RST and TFT_BLK loops is absolutely authorized.Your proposed fallback mapping is flawless. We will officially freeze the telemetry lines on your suggested free ADC pins:TI AMC3301 Current Telemetry Output → PA2 (ADC1_IN2)TI AMC3311 Voltage Telemetry Output → PA3 (ADC1_IN3)This completely resolves the HMI display collision and leaves your existing reset and backlight layouts untouched. I have updated our master pin table on this end to ensure the firmware registers synchronize perfectly with your physical **traces**.You are completely clear to lock the MCU sheet and push forward!Best Respectfully,Dave

DK

Dave Kreutzman11:19 PM

Subject: URGENT: PCBA Layout Schematic Additions — Missing MCU Logic PinsHi Febryan,To ensure our physical copper trace layout perfectly coordinates with our synchronized backend C++ firmware compilation loops, please apply these final three logic pin allocations directly to your STM32F407VGT6 microcontroller schematic sheet before freezing your routing nets for Friday's deliverable:Capacitive Touch Interface Controls (FT6336G):Route the Touch Interrupt Line (Touch INT) → Pin PB0 (This must be mapped cleanly back to the MCU to handle low-power tap-to-wake interrupt loops).Route the Touch Reset Line (Touch RST) → Pin PB1.Onboard Memory Bus (24C256 EEPROM):Confirm that the I2C lines for the EEPROM storage module sit directly on the same I2C bus footprints (Pin PB6 for SCL / Pin PB7 for SDA) as the touch controller. It will run on an independent memory address, completely eliminating the need for a secondary physical I2C bus track on the board canvas.External Auxiliary Dry Contact Port (AUX Relay):Route the low-voltage logic gate control line for the generic addressable AUX

relay circuit directly to Pin PD11. This groups its logic trace neatly alongside your primary high-power mechanical relays on Port D (PD12–PD15). Please confirm that these three lines are successfully injected into your active design sheets, and keep your 48-hour progress notes flowing! Respectfully, Dave

Please confirm receipt of above transmission.

Tuesday, Sep 01

DK

Dave Kreutzman 4:06 AM

Subject: Schematic Amendment — Dual-Input Parasitic Logic Supply & RTC Domain

Hi Febryan, To guarantee true 24/7/365 system uptime regardless of whether Solar is absent or the AC Grid is disconnected, please implement an isolated dual-input parasitic power bus to feed our central low-voltage logic plane:

AC-Side Logic Tap: Embed a miniature, board-mounted, high-efficiency encapsulated AC-DC converter module (such as a potted Mornsun LD05-23B series) wired parasitic-style directly across the incoming 240V AC nominal utility grid lines prior to any mechanical load relay footprints.

Zero-Heat Power Path Sequencer: Tie the low-voltage DC outputs of both the wide-input PV Mornsun module and your new auxiliary grid AC-DC module onto a shared logic rail using an Ideal Diode / Power Path Controller Chip layout (such as an LTC4412 or LM5050 framework). This enforces a perfect, zero-heat, time-sequenced power handoff within our sealed enclosure while keeping the Solar ground and Grid ground entirely isolated.

RTC Backup Battery Realignment: Utilize a standard, non-rechargeable CR2032 coin cell battery wired strictly to the STM32's VBAT pin to protect the hardware clock during a total power failure. Place a standard 0.1µF decoupling capacitor right next to the holder clip and ensure zero GPIO pins are left floating in the backup domain.

Physical Accessibility: Position the horizontal coin-cell holder clip at the absolute outer bottom edge of the low-voltage logic plane, a minimum of 2.0 inches clear of all high-voltage traces, allowing a technician to safely slide the cell out without tool intervention.

Best regards, Dave

Please confirm receipt.

Feb Riz 9:13 AM

Hi Dave,

Received both thanks. Adding these now: AUX relay → PD11, RTC (CR2032 → VBAT, 0.1µF, edge holder), and confirming EEPROM + touch already share I2C1 (PB6/PB7) no change needed there.

Two things before I commit the rest:

Touch INT/RST → PB0/PB1: these currently sit on PC2/PC3. Confirming this is a move (PC2/PC3 released), not a second set, so touch isn't double-mapped?

Power path: an ideal-diode OR (LTC4412/LM5050) merges both supplies onto one rail, which

makes their grounds common at that point which conflicts with "keep Solar and Grid grounds fully isolated." We can have one OR'd rail (common ground) or fully isolated grounds, not both. Which do you want? If isolation is the priority, the AC-side supply likely needs its own isolated DC-DC into the shared rail. Also please confirm the Mornsun LD05-23B PN is rated for 240 VAC; I'll add the required AC fusing/creepage.

Given the AC supply + power-path is a sizable block landing near Friday, let me know if it lands in M2 or the next phase so it gets proper rigor.

Confirm the two above and I'll inject everything. 48-hour notes coming.

Respectfully,

Febryan

DK

Dave Kreutzman 9:58 PM

Subject: Update: Technical Audit Active — Hold on Schematic Freeze Hi Febryan, Thank you for your response. Your questions regarding the touch pin double-mapping and the grounding/isolation boundaries on the dual-input power path are exceptionally sharp, and they have triggered an internal technical audit on our end this morning. We are currently running a thorough engineering review across our master specifications to lock down the exact structural parameters for the auxiliary AC supply layer, global pre-compliance safeguards, and serviceability components. Please place a temporary hold on freezing your MCU sheets or routing any copper traces for the auxiliary power block right now. I am in an executive business session this morning, but I will return to your workspace chat box later this afternoon with our finalized, synchronized directives and an adjusted milestone timeline to ensure you have ample, un-rushed runway to execute this block with the proper **rigor**. Talk soon, and thank you for your exceptional engineering diligence! Best regards, Dave

Wednesday, Sep 02

Feb Riz 7:43 AM

Understood, I'm placing a hold on the auxiliary power block, the pre-compliance/surge safeguards, and the related MCU pins until your finalized directives this afternoon. No copper routing on those, as requested.

In the meantime I'll keep momentum on the parts unaffected by the audit: finishing the chopper power stage (MOSFET/diode/bus caps done, adding the MOSFET snubber), verifying the completed isolated sensing stage, and cleaning up ERC. I'll hold the MCU sheet freeze until your word.

Looking forward to the updated directives and timeline. Thanks for the thorough review — happy to build this block properly rather than rushed.

Respectfully,
Febryan

DK

Dave Kreutzman 9:21 PM

Subject: Final Technical Directives: Star Power Hub Architecture & Timeline Realignment

Hi Febryan, Thank you for your exceptional engineering diligence and for maintaining rigorous momentum on the copper chopper stages while we finalized this technical audit. Our internal design review is complete. To achieve a 10-year, near-military-spec outdoor durability target while protecting our global standalone solar markets and universal grid-backup commercial pipeline configurations, we are executing a major subtractive engineering refinement. We are completely eliminating the active time-sequencing integrated circuits (ICs) and switching transistors from the power path. Please implement the following definitive architectural specifications directly into your sheets:

The Touch Interface Pins: Confirmed. This is a permanent move, not a double mapping. Please completely release pins PC2 and PC3 back to the free GPIO pool, and permanently map the touch controller lines strictly to PB0 (Touch INT) and PB1 (Touch RST) to match our firmware registers.

The "Star Power Hub" Galvanic Ground Solution: You are completely correct about circuit physics: merging two low-voltage DC rails onto a shared bus forces their grounds to be common at that exact junction. To maintain our strict mandate that the external high-voltage Rooftop PV strings and the raw 240V AC Utility Grid lines remain 100% galvanically isolated from each other, we are utilizing direct internal transformer barriers. Because both your wide-input PV Mornsun module and the new Mornsun auxiliary AC module feature native internal galvanic transformers, their low-voltage DC outputs are fully authorized to merge onto a Common Safe Logic Ground Plane to run the STM32 chip and HMI display screen. The raw external inputs remain perfectly isolated by air gaps and magnetic fields in the real world.

Passive Diode "OR" Power Path: Completely delete the active ideal-diode sequencer chip frameworks (LTC4412/LM5050) and external logic switching MOSFETs. Instead, merge the low-voltage output legs of the PV Mornsun and the auxiliary AC Mornsun together onto the central logic bus using a passive, back-to-back Schottky Blocking Diode "OR" gate layout. This uses pure physics to automatically select the active power source with absolute zero thermal generation or complex code dependencies.

Consumable Elimination — External Panel Breaker Button: Do not use internal glass fuses on the auxiliary AC input line. Instead, integrate a tiny, board-routed Panel-Mounted Resettable Thermal Circuit Breaker Button (0.5A or 1A rating, such as a TE Connectivity W28 series or direct equivalent). This component must mount to the edge of the board so it protrudes through a tiny cutout on the safe bottom-edge wall of the polycarbonate box, right next to our waterproof cable strain-relief glands, allowing zero-tool external user serviceability if a grid surge hits the home. Ensure the high-voltage internal terminals of the breaker button are safely shrouded within your high-voltage layout zone to

satisfy UL/CE physical insulation requirements. Please enforce standard slow-blow characteristics and strict IPC-compliant creepage/clearance trace slots around this high-voltage AC landing header.

Desert-Shield RTC Battery Upgrade: To ensure our inventory can withstand long-term warehouse container storage in extreme summer desert climates without accelerated chemical self-discharge, upgrade the coin battery specification from a consumer CR2032 to an industrial, automotive-grade Panasonic BR2032 Poly-Carbon Monofluoride Lithium Battery [source: 1.3.2, 1.3.8]. It drops directly into your existing horizontal, edge-mounted SMD coin cell retaining clip footprint with zero layout modifications [source: 1.2.1, 1.3.2].

Milestone 2 Timeline Extension: To ensure you have ample, un-rushed runway to execute these high-voltage clearances and structural components with pristine trace precision, your Milestone 2 data drop deadline is officially extended to Wednesday, September 9, 2026. You are completely unblocked to freeze the microcontroller sheets and commence full copper board trace layout routing!

Respectfully, Dave

Thursday, Sep 03

Feb Riz6:58 AM

Hi Dave,

Received, and thank you the directives are clear and the extended timeline is appreciated. The passive Schottky-OR approach is a clean simplification, and your point on the isolation is exactly right: since both Mornsun modules carry their own internal transformer barriers, merging their DC output legs onto a common logic ground is fully valid while the external PV and AC inputs stay galvanically isolated. Good resolution.

I'm proceeding on the confirmed items now: touch pins moved permanently to PB0/PB1 (releasing PC2/PC3), the BR2032 RTC upgrade, and building the auxiliary block in schematic (AC-side Mornsun → passive back-to-back Schottky OR → common logic bus, plus the resettable breaker on the AC input).

Two quick confirmations before I commit the high-voltage parts:

Auxiliary AC module PN: which specific Mornsun AC-DC part are you specifying for the 240 VAC input? I want to lock the exact voltage/current rating and footprint rather than assume, given it's a mains-referenced stage.

Milestone 2 scope: to plan the runway correctly does "full copper trace routing" mean M2 now delivers complete PCB layout + routing across all domains (including the new HV-AC block, TO-247 chopper, and breaker landing), not just placement + STEP? Just aligning expectations so I sequence it properly: I'll freeze and ERC-clean the full schematic first, then route.

I'll treat the AC breaker landing and creepage/clearance zones with the UL/CE rigor you outlined. Progress notes continuing every 48 hours.

Respectfully,
Febryan

DK

Dave Kreutzman 7:50 AM

Hi Febryan, Thank you for the quick turnaround. Your schematic freeze progress on the high-power copper chopper stages looks excellent. Please lock these two definitive parameters straight into your board design: Auxiliary AC Module Part Number: Please utilize the Mornsun LD10-23B05R2 module footprint for the mains-referenced tap. We are specifying the 10-Watt model to guarantee ample, stable current headroom to power the central +5V common logic plane, the touchscreen controller, and the Tuya module tasks without thermal strain. Milestone 2 Routing Scope Confirmation: Confirmed. Once you clear ERC and freeze the schematics, your Wednesday, September 9th Milestone 2 delivery requires the full PCB layout and complete copper trace routing across all domains (including the new HV-AC block, TO-247 chopper path, and shrouded resettable thermal breaker landing). Enforce strict IPC creepage and clearance slots around that AC header to guarantee our smooth pass through UL/CE testing. Looking forward to your next progress drop! Best regards, Dave

Saturday, Sep 05

DK

Dave Kreutzman 5:44 AM

Subject: LSS-100 Portrait Rev 3.9: Relay Pin Map Verification & Hardware Freeze Febryan, as we head toward the September 9th M2 copper layout freeze, we need to formally cross-check your physical PCB trace routing against the firmware bench layout. Anand is currently writing the driver code, and we must ensure there are zero pin collisions or backwards contactor triggers on first power-up. Anand's current bench code assumes the following pin configurations on the STM32 chip (PD12 through PD15): RLY1 (PD12): Solar Contactor RLY2 (PD13): Utility Contactor RLY3 (PD14): Unused / Reserved RLY4 (PD15): Load Follower / Master Supervisor Please cross-examine this map against your active copper traces. If your layout routes these differently, please reply with your exact trace map immediately so we can adjust the firmware side before the freeze. Additionally, please verify that your layout incorporates the following frozen specifications from the Portrait Rev 3.9 / LSS-100 Master Specification: The Star Power Hub Logic Plane: No active switching transistors. Verify that the wide-input Mornsun PV (+5V) and auxiliary Mornsun AC-DC low-voltage (+5V) logic output rails merge onto the shared plane via the passive back-to-back Schottky Blocking Diode "OR" gate configuration. Optocoupled Ground Isolation: Ensure Mikrobus optocoupled ground isolation is fully enforced across all mechanical relay pins. Consumable Elimination: Confirm that internal

glass fuse brackets are deleted and replaced by your board-routed, external-facing TE Connectivity W28 Series Panel-Mounted Resettable Thermal Circuit Breaker Button. RTC Protection: Confirm the footprint is committed to the automotive-grade Panasonic BR2032 Poly-Carbon Monofluoride Lithium Battery in your horizontal edge-mounted clip. Please send over your trace layout confirmation as soon as possible so we can give Anand the green light on the pin mapping. Best regards, Dave

DK

Dave Kreutzman 11:46 PM

Febryan, To guarantee absolute alignment ahead of your Wednesday, September 9th Milestone 2 full copper trace routing data drop, we have finalized the exact physical dimensions, substrate constraints, and trace widths on our desktop records. Please lock these definitive IPC Class 3 engineering parameters directly into your active board layout immediately:

- Copper Thickness & Power Trace Parameters (3 oz Core)** To achieve our 10-year outdoor durability target under an absolute worst-case ambient container storage profile of 150°F, you must enforce heavy-duty thermal paths:
Copper Weight: Enforce a minimum base copper thickness of 3 oz/ft² (105 µm finished) on all high-voltage/high-current power routing layers.
Power Trace Widths: All continuous 30A current loops (AC Utility and DC Solar lines) must maintain a minimum external layer width of 5.1 mm, safety-factored up to a standard 10.0 mm wherever routing layout permits.
Creepage Isolation: Maintain a strict physical creepage and clearance gap of at least 6.3 mm (0.25") between any high-voltage tracks and the low-voltage Common Safe Logic Ground Plane.
- Mechanical Enclosure & Edge Notching Constraints** The motherboard PCB will be flush-mounted directly against the flat face of our robust aluminum flatback profile block inside the vertical Fibox PC 25/22-FC3 portrait housing. Please commit the following virtual board canvas layout limits to your schematic sheet:
Maximum Board Outline: 242 mm Height × 205 mm Width max. This guarantees clean safety clearance margins against the enclosure walls.
The Midpoint Edge Notch: The enclosure features an internal molded divider shelf at its horizontal centerline. At exactly the 121 mm vertical height mark of your board outline, you must notch/radius the vertical fiberglass outer edges of the PCB inward by 4 mm on each side across a 15mm vertical band.
Midpoint Keep-Out Band: Keep all power traces, logic lines, and signal vias at least 6 mm away from the outer vertical board edges along that midpoint band. This ensures our traces are completely protected when the factory CNC routes the pass-through slot.
Component Height Ceiling: High-profile items (such as the encapsulated Mornsun LD10 and PV15 isolation modules) must remain under a maximum component height ceiling of 35 mm from the top face of the PCB to ensure they sit safely below the upper enclosure wall.
- External Component Alignments**
Resettable Breaker Button: Place the footprint for the panel-mounted TE Connectivity W28 button flush along the absolute bottom-center edge of the board outline. It must extend downward past the board edge by 5mm to slide cleanly through our lower

enclosure wall cutout. HMI Display Alignment: Center the 3.2-inch capacitive touch display module on the top half of the board so its active viewing area lines up perfectly underneath our centralized 85mm Width × 55mm Height landscape cover window.

4. Definitive Relay Pin Mapping (PD12–PD15)

Double-check that your copper paths to the STM32 microcontroller match Anand's exact bench assumptions to prevent cross-contactor triggers on power-up:

RLY1 (PD12): Solar Contactor
RLY2 (PD13): Utility Contactor
RLY3 (PD14): Unused / Reserved
RLY4 (PD15): Load Follower / Master Supervisor

Please provide your formal status update and confirmation of these integrated 3 oz layout constraints within the next 48 [hours](#). [Best](#) regards, Dave

[hours.best](#)

GROUP

Dave Kreutzman 8:26 PM

Hi Feb and Anand,

I have created this joint engineering room so we can coordinate our hardware-to-software pin mapping in real time.

Feb is leading our physical 3 oz. copper board substrate layout pass, and Anand is spearheading our custom C++ firmware architecture for the Alset 100 platform.

Feb, please deliver our target schematic sheets, STM32F4 part numbers, EEPROM data, and ADC sensor calibration channels directly into this thread the moment you establish our layout foundation this week so Anand can map his code drivers against our exact physical traces.

Let's build this together.

Feb Riz 8:29 PM

Thanks, Dave, and hello Anand.

I'll send the schematic soon; it's almost ready, though it still needs some time for a final review.

Anand Kumar N S 8:36 PM

Hi Feb, thanks for jumping in — good to be working directly on the pin mapping.

No rush on the schematic, appreciate you doing a proper final review before sending. Whenever it's ready, the pieces that'll let me map drivers cleanly against your traces are:

1. STM32F4 part number (matters for flash size / RDP2 config)
2. Pin assignments for the TFT/touch SPI bus, ADC channels (temp probe + current shunt), relay driver GPIOs, and the leak-sensor input
3. Touch controller IC and TFT/LCD controller IC part numbers
4. EEPROM part number + interface (I2C?)
5. NTC probe part number or its resistance-vs-temperature table
6. Tuya WB3S interface details (UART/AT commands?)

In the meantime I'm not blocked — building and testing Milestone 1 (touchscreen UI, wizard, power filters) on my own STM32F4 + SPI touchscreen testbench with placeholder pins, so it'll be a clean swap once your traces lock. Happy to hop on a quick call if that's faster for walking through the pin map together.

— Anand

DK

Dave Kreutzman 8:55 PM

Gentlemen, We are officially pushing down a critical executive pivot regarding our final chassis design track. Please review and lock these five matching design criteria into your board canvas boundary layouts and firmware UI layers starting this morning:

Material Architecture: We are officially scrapping the legacy stamped sheet metal enclosure footprint. The Alset 100 platform will utilize a high-sophistication, domestic US-sourced Molded Opaque Grey Polycarbonate NEMA 3R Enclosure baseline (UL94-V0 flame-retardant and UV-stabilized).

Chassis Target Dimensions: Our primary engineering envelope is locked to a standardized commercial catalog footprint measuring approximately 200mm Height × 120mm Width × 75mm Depth (approx. 8" × 4.7" × 3") with a solid, non-transparent lid. This includes standard molded-in internal mounting bosses and robust high-torque machine screw inserts to prevent field stripping.

Passive Radiator Heatsink Integration: Because polycarbonate acts as a thermal insulator, our deep-finned aluminum cooling heatsink will pass straight through an automated CNC-routed rectangular cutout window on the flat back wall of the plastic box. High-power transistors must align along this rear boundary to mechanically mate directly to the exterior cooling fins. A compressed tongue-and-groove neoprene or silicone gasket will seal this rear frame to pass NEMA 3R rain-jet standards.

100% Internal Antenna Topology: Because polycarbonate is entirely RF-transparent, do not route any antenna wires or connectors outside of the enclosure. The Tuya WB3S module's trace antenna will remain completely protected inside the unbroken side wall of the plastic housing. Feb, please place the Tuya footprint directly against the upper vertical side-edge corner of the PCB layout canvas, maintaining a strict copper-free Keep-Out Zone behind it to guarantee full 2.4 GHz signal punch straight through the plastic to the building router.

EMI Suppression Core: To counteract the lack of natural metal EMI shielding and pass FCC emissions tests effortlessly, Feb will incorporate localized ground isolation planes right into the multi-layer 3 oz. copper layout substrate to filter any 20 kHz chopping noise directly at the board level. Feb, please source the official 3D .step file and internal mounting post configurations for this standard container class today so your physical PCB boundaries snap seamlessly into the factory posts. Anand, coordinate your HMI touchscreen interface parameters directly against these dimensions. Thank you. Dave

Anand Kumar N S9:29 PM

Noted, Dave - for reference, my dev testbench is running a small color TFT on the ILI9341 driver (240x320 resolution, 2.4-2.8" class panel, the common size for this controller). That's a stand-in for validation only, not a production recommendation. Once Feb has the actual target display's part number and physical cutout size, I'll size the wizard/dashboard UI layout against those real numbers. No blocker on my end for now - testbench work continues in parallel.

Feb Riz11:05 PM

Hey Anand — good to be coordinating directly on this. Quick status on the pin map: part of it's locked, a couple of pieces still need decisions before I can hand you final numbers.

Locked (exact pin numbers to follow once we lock the F4 variant — see below):

1. Analog sensing — NTC temp probe and current shunt each on their own ADC channel, both buffered through an LM358 front-end.
2. Leak sensor — on an ADC-capable input, so you can read it digital or analog and set your own threshold (water resistance varies a lot).
3. Relay drivers — 4 GPIOs driving the transfer-switch coils (RLY1–4).
4. EEPROM — 24C256 over I2C.
5. Tuya WB3S — UART (Tuya MCU serial protocol).

TFT + touch: good that you flagged this. Our earlier draft assumed a UART smart-display, so I'm reworking the MCU sheet into a proper SPI TFT bus (SCK/MOSI/MISO/CS/DC/RST + backlight) plus the touch lines. One gap, though: the production display and touch controller aren't picked yet — your ILI9341 is a testbench stand-in, as you noted. Before I can give you real controller part numbers and the physical cutout, we need to lock the production panel. Can we get Dave to confirm who owns that call? Once it's chosen I'll match the SPI/touch pin map to it.

F4 part number: we'd been on an F103, moving to F4 to line up with your firmware. Since the UI assets are flash-hungry, I'd lean toward a 1 MB-flash variant (e.g. STM32F407VGT6) for headroom — but you know your UI footprint best, so tell me your flash/RAM target and I'll lock the exact variant. That also settles your RDP2 question.

Still open: NTC probe part number / R-T table — I need that to finalize the divider values and your temp conversion. Do we have a chosen probe, or should I propose one?

Net: I'll send the final F4-based pin map the moment we lock (a) the F4 variant and (b) the production display. Nothing here should block your Milestone 1. And yes — happy to jump on a quick call to walk the pin map together, probably faster than the thread. Just say when

Tuesday, Aug 18

DK

Dave Kreutzman 1:07 AM

Gentlemen, Excellent synchronization here. Let's freeze our production panel specifications immediately so Feb can lock the SPI pin-outs and screen cutout boundaries on our 200mm × 120mm polycarbonate frame today: The Display Panel Target (The Global Workhorse Baseline): Size Class: We are standardizing on a 3.2-inch Diagonal color TFT LCD module. This provides an elite elongated aspect ratio that fits our vertical polymer frame beautifully, leaving a sophisticated bezel margin around the window. Resolution: Standard 240 × 320 Pixels (RGB Vertical Stripe). Core Display Controller IC: Strictly specify the ILI9341 driver. This completely lines up with Anand's testbench firmware libraries, allows clean 4-pin/5-pin SPI serial communication to save pins, and ensures we can source from dozens of interchangeable

drop-in global vendors instantly. The Capacitive Touch Controller IC: Standardize the capacitive glass overlay on the FocalTech FT6336G or the XPT2046 controller IC mapped over an I2C/SPI bus. This eliminates manual screen calibration entirely, allows true 'tap-to-wake' register sensing in the dark, and is massively mass-produced for high cost-efficiency. The Microcontroller Variant Core (Feb's Lean): Feb, I completely agree with your assessment. Let's officially lock the main brain onto the STM32F407VGT6 in the LQFP100 package. The 1 MB of internal Flash memory and 192 KB of RAM gives Anand massive headroom to bake in all five language text string matrices, our dynamic 4-color status icons, your RDP Level 2 security flash-locking script, and future over-the-air (OTA) Tuya update payloads without ever hitting a memory bottleneck! The Temperature Probe Baseline (Dave's Fixed Rule): We are strictly standardizing on our field-proven 5K Ohm NTC Thermistor Probe with a strict $\pm 1\%$ tolerance. Feb, use a standard 5K Ohm curve matrix to finalize your resistor voltage-divider parameters and LM358 operational amplifier front-end gain values today. Gentlemen, please freeze the schematic around these four specific targets. Feel free to jump on that quick pin-mapping coordination call this afternoon to lock down the exact STM32F4 registers.

DK

Dave Kreutzman 3:17 AM

Feb — To follow up on our chassis update, we are officially freezing the rear heatsink configuration.

Do not design a custom profile or include a custom extrusion tooling budget for this launch. We are standardizing strictly on an open-source, standard off-the-shelf 4.850" wide aluminum profile rail (Alloy 6063-T5, approximately 0.14" base thickness with standard cooling fins).

Please pull the standard engineering templates for this 4.850" catalog profile class today. Size your board's rear component layouts and screw mounting coordinates so our high-power transistors bolt straight onto this factory footprint through Vergent's back cutout window.

Anand, this freezes our thermal mass baseline for your code calculations.

Thank you,
Dave

DK

Dave Kreutzman 5:43 AM

Feb —

Rethinking our chassis parameters, we want to look at our rear heatsink geometry and mounting layout based on your upcoming thermal diagnosis.

Please evaluate a heavy-duty, standard off-the-shelf 4.850" wide catalog aluminum extrusion profile class featuring a thick base plate (targeting a ~0.312" / 5/16" floor baseline) with vertical cooling fins (not the thin base specified by me in error earlier) to handle our 20 kHz solid-state chopper heat dissipation requirements. Do not utilize an extrusion profile that relies on continuous molded side legs on the upper face, as that will interfere with a flat cutout window on the back wall of our polycarbonate box.

Instead, look at a direct-mounting setup: The manufacturing house will run a fast drilling and countersink milling operation right through the face of the thick aluminum back floor. We will use flat-head countersunk machine screws passing through the face to lock the radiator tightly to the rear polycarbonate frame. This keeps the screw heads completely flush, ensuring zero drag or turbulence inside our vertical columnar airflow channels, while providing high-torque compression against our internal rear NEMA 3R gasketing.

Two open items I need your specialized engineering feedback on:

1. Thermal Engineering Calls: Please run the full thermal dissipation and heat-mapping simulations for our high-power transistors against this heavy-base 4.850" catalog vertical rail baseline. Let me know if this surface area gives us the necessary cooling safety factor, or if we need to deepen the fins or extend the block vertically past the top roofline of the plastic box to act as an un-blockable safety chimney.

2. Customer Wall Mounting & Standoffs: We need to establish how the final customer will physically mount this entire assembled controller box to a wall in the field, keeping in mind that the heatsink MUST sit completely off the mounting surface to allow proper vertical chimney airflow underneath.

Please audit the industrial catalogs to see if there is a stock 4.850" profile featuring integrated bottom stand-off rails/legs (similar to our legacy setups) to lift the fins off the wall natively, or analyze using standard stock flanged end-cap channels bolted to the block to act as our wall mounts. Let me know the best structural layout so the unit carries its weight safely without putting mechanical stress on our plastic enclosure walls or breaking our NEMA 3R weather seal.

Please run the thermal calls and mechanical review on this layout baseline and let me know your thoughts once your schematic review is ready to drop.

Thank you,

Dave

DK

Dave Kreutzman 5:55 AM

Hello Feb, To close out my layout input for this phase and protect our development velocity, here is our final mechanical manufacturing specification for the rear radiator: We will utilize a blind-drill and tap configuration machined directly into the inside face of our heavy-duty 0.312" aluminum base plate. Fasteners will mount from the inside of the polycarbonate box facing outward. This completely eliminates any through-drilling to the outside world, guaranteeing a 100% impenetrable NEMA 3R exterior face that protects our internal components from any driving rain or moisture entry points. Moving forward, I am officially stepping away from the design track to let you execute. I am paying for your high-level hardware expertise, so I want you to take ownership of these parameters, run the necessary thermal/mechanical calls, and finalize our pin-mappings as a unified team with Anand. Do your thing, run the math, and protect our engineering integrity. I will stand down and stay completely out of the thread until you are ready to drop your finalized schematic sheets and pin-out sheets for my executive review. Let's see what you've got.

Respectfully, Dave

Wednesday, Aug 19

DK

Dave Kreutzman 5:04 AM

Hi Feb, Got it, I see it's a compressed .rar archive now. Since I am reviewing our engineering tracks on my field iPad and Mac dashboard without full CAD software loaded, I can't open raw project archives on this screen. To help me run my executive review, verify the layout, and release your funds immediately, please reply directly into this thread with two uncompressed, standard files: A Universal PDF of your schematic sheets (so I can review the 5K NTC and relay traces). A 3D .step file or a simple high-res image showing the physical board outline and your blind-tapped mounting holes against our 4.850" heavy aluminum heatsink baseline. The moment you drop those quick viewable sheets into our chat box so I can verify the layout parameters, I will immediately jump onto my dashboard and release this milestone payment to you! Thanks, Dave

DK

Dave Kreutzman 5:16 AM

Hey Riz, I was just able to remotely log into my work Dell. got it opened. Reviewing now. Thx.

DK

Dave Kreutzman 6:08 AM

Feb —

I have expeditiously completed a thorough, line-by-line executive and mechanical engineering audit across our entire project framework—including the BOM spreadsheet, the MCU pin maps, the Isolation lines, the Sensing sheets, and the Switching/Power circuits.

This current package is an early architectural draft containing critical engineering gaps and temporary placeholders. I will not authorize an escrow payment release until the core high-voltage engineering, noise isolation, and discrete power paths are fully designed and drawn onto our KiCad sheets.

Please halt layout mapping and correct these four primary deficiencies immediately:

1. Discrete Power Deck & Isolated Gate Supply (Power Sheet & Section 3.1 / 3.3): Using pre-assembled plug-in 'Buck Modules' (24->12V and 12->5V) is unacceptable for our production timeline. These placeholders must be scrapped. Please design these step-down regulators directly onto our primary board canvas using standard, discrete silicon ICs to allow fully automated high-speed SMT factory assembly. Furthermore, our isolated gate supply (net VG+) for the high-speed TLP250 / HCPL-3120 gate drivers cannot be left empty; you must fully engineer and draw these isolated power traces on this pass.

2. Isolated PV Voltage Sensing & 70V Cutoff (Isolation/Power Sheets & Section 3.2 / 4.1): To be clear, yes—the MCU absolutely must read incoming PV bus voltage so our software can calculate live solar wattage columns and execute our California Title 24 overrides. Replace the PC817 placeholder on the isolation sheet with a proper isolated linear optocoupler network (utilizing an IL300 or isolated amplifier) to bridge the barrier safely. Regarding the Mornsun 100V under-voltage limit, do not accept the lockout drift; incorporate a low-cost high-voltage startup resistor loop or auxiliary path so our STM32 logic can remain safely energized down to our 70V solar threshold.

3. Wireless Noise & Port Isolation (MCU Sheet Pin Map): Routing our sensitive Tuya WB3S Wi-Fi/Bluetooth UART lines onto Pins PA2 and PA3 (USART2) places them directly adjacent to our high-voltage analog inputs and right beneath our high-frequency 20 kHz solid-state chopper on Pin PA8. The intense switching noise will violently corrupt our wireless data packets. Please move the Tuya communication network completely off Port A and re-route it to an isolated serial port on the opposite side of the chip—specifically USART3 on Port C (Pins PC10 & PC11)—to guarantee total wireless data immunity and flawless iPad screen mirroring.

4. EMI Filtering Integration (Section 5.1): Housing a kilowatt-class 20 kHz chopper inside a non-metallic polycarbonate enclosure requires strict emissions management to pass FCC Part 15 Class B testing. Your schematic must explicitly incorporate our multi-layer common-mode chokes and X/Y filtering capacitor network right now. We will not treat EMI as an out-of-scope specialist task later on.

Please take ownership of these high-power parameters, run your thermal calls against our locked-in heavy-duty 4.850" vertical catalog heatsink baseline, and draw the true power silicon onto these sheets. I am standing down and staying completely out of the thread to let you execute. I look forward to reviewing a finalized, zero-error production schematic package once your full team review is ready to drop.

Let's build this correctly.

Respectfully,

Dave

Monday, Aug 24

Feb Riz8:54 PM

Hi Dave,

Message received loud and clear. I appreciate the rigorous executive audit. You are absolutely right, these high-voltage and isolation parameters need to be hardcoded into the architecture right now, not treated as later additions.

1. Discrete Power Deck & Isolated Gate Supply: I am scrapping the plug-in buck modules entirely. I am designing discrete, factory-ready SMT step-down regulators (24V to 12V, and 12V to 5V) directly onto the primary canvas. I am also fully engineering the isolated gate supply (VG+) circuit to drive the TLP250/HCPL-3120 optocouplers.

2. PV Sensing & 70V Auxiliary Loop: The PC817 is out. I am integrating a proper isolated linear optocoupler/amplifier network to safely bridge the barrier for accurate PV bus voltage telemetry. To solve the Mornsun 100V UVLO limitation, I am adding a high-voltage auxiliary startup circuit to keep the STM32 logic and screen safely energized down to our 70V threshold.

3. Wireless Noise Isolation: I am completely moving the Tuya WB3S UART lines away from the 20 kHz chopper noise on Port A. They are being re-routed to the isolated USART3 on Port C (Pins PC10 & PC11) to guarantee zero data corruption. I will update the pin map for Anand as soon as this is locked.

4. EMI Filtering: EMI filtering is no longer out-of-scope. I am drawing the multi-layer common-mode chokes and X/Y filtering capacitor networks directly onto the power sheet right now to ensure we hit FCC Part 15 Class B compliance inside the polycarbonate chassis.

I am taking full ownership of these power parameters and will drop the finalized, zero-error production schematic for your review shortly so we can unlock Milestone 1 and push straight into the 3 oz. heavy copper layout pass.

I'll fully lock in on hitting the August 31st deadline for the Fast-Track Gerbers. I will update you as soon as the revised sheets are ready.

Best regards,

Febryan